



MANIPAL

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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Cryptology – 4_Theory

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Assignment 4 - Theory

1) Cipher text: HLD TGIH JDXW QCMAG using Hill cipher

$$\text{key} = \begin{bmatrix} 3 & 2 \\ 5 & 1 \end{bmatrix}$$

Use (A=0, B=1, ..., Z=25) Find plain text

$$\det(k) = 3(1) - 2 \times 5 = 11$$

~~$11^{-1} = 19 \pmod{26}$~~ $\Rightarrow 26 = 2(11) + 4 \rightarrow (3)$
 $11 = 2(4) + 3 \rightarrow (2)$
 $4 = 1(3) + 1 \rightarrow (1)$
 $3 = 3(1) + 0 \quad \times$

$$4 = 1(3) + 1$$

$$1 = 4 - 3$$

$$11 = 2(4) + 3$$

$$1 = 4 - [(11) - 2(4)]$$

$$1 = 3(4) - 11$$

$$4 = 26 - 2(11)$$

$$1 = 3(26 - 2(11)) - 11$$

$$1 = 3(26) - 6(11) - 11$$

$$1 = 3(26) - 7(11)$$

$$\therefore 1 = (-7)(11) + 3(26)$$

$$-7(11) \equiv 1 \pmod{26}$$

$$11^{-1} \equiv -7 \pmod{26}$$

$$\therefore 11^{-1} \equiv 19 \pmod{26} \Rightarrow 11 \times 19 = 209$$

$$\underline{11^{-1} = 19}$$

The adj of k is

$$\text{adj}(k) = \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$k^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

$$k^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

Divide the ciphertext into pairs

$$\begin{array}{ccccccc} \text{HL} & \text{DT} & \text{QI} & \text{HJ} & \text{DX} & \text{WQ} & \text{CM} & \text{AG} \\ (7, 11) & (3, 19) & (16, 8) & (7, 9) & (3, 23) & (22, 14) & (2, 12) & (0, 6) \end{array}$$

For each pair

$$P = k^{-1} C \pmod{26}$$

$$\text{HL} \Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} = \begin{bmatrix} 175 \\ 118 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$$

$$19 = T, 14 = O$$

$$PT = TO$$

$$\text{DT} \Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} = \begin{bmatrix} 275 \\ 122 \end{bmatrix} \pmod{26} = \begin{bmatrix} 15 \\ 18 \end{bmatrix}$$

$$15 = P, 18 = S$$

$$PS$$

Q1

$$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} = \begin{bmatrix} 160 \\ 184 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

Plain = EC.

HJ

$$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} = \begin{bmatrix} 147 \\ 108 \end{bmatrix} \pmod{26} = \begin{bmatrix} 17 \\ 4 \end{bmatrix}$$

Plaintext = RE

DX

$$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} = \begin{bmatrix} 290 \\ 278 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 18 \end{bmatrix}$$

Plaintext = SA

CM

$$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} = \begin{bmatrix} 174 \\ 78 \end{bmatrix} \pmod{26} = \begin{bmatrix} 18 \\ 0 \end{bmatrix}$$

Plaintext = GE

Combining the decrypted pairs:

TOPSECRET MESSAGE

2. Consider the HILL Cipher key matrix

$$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$$

Cipher text : RXAODY

Find the plain text for the above cipher text

Decryption:

$$P = (K^{-1} \times C) \bmod 26$$

$$\begin{aligned} \text{Det}(K) &= (24 \times 14) - (19 \times 5) \\ &= 336 - 95 = 241 \equiv 7 \end{aligned}$$

7 is a coprime with 26
 $\Rightarrow 7^{-1} = 15$

$$\Rightarrow K^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} = \begin{bmatrix} 210 & 605 \\ -75 & 360 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

$$RX = \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} \bmod 26 = \begin{bmatrix} 57 \\ 557 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix} \Rightarrow \begin{matrix} F \\ L \end{matrix}$$

$$AO = \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} \bmod 26 = \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{matrix} O \\ W \end{matrix}$$

$$DY = \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} \bmod 26 = \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \begin{matrix} E \\ R \end{matrix}$$

Plain text $\Rightarrow$ FLOWER

3. Confusion vs. Diffusion

Analyse the following classical ciphers with respect to confusion and diffusion. Complete the table and justify your classification based on how each cipher transforms the plaintext.

Cipher	Does it change the character/symbol value?	Does it change the character position?	Confusion / Diffusion	Justification
Shift cipher	Yes	No	Confusion	Substituted via mathematical shift ($p+k$). Character order is unaltered
Multiplicative Cipher	Yes	No	Confusion	Substituted via modular multiplication ($p \times k$). Character order is unaltered
Affine Cipher	No	Yes	Diffusion	Characters retain the original value, but their positions are scrambled across the text.

Transposition Cipher	Yes	No	Confusion	Multiplicative & shift structures ((a-p+b) mod 26) position remains unchanged
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Relationship between substitution and confusion:

In a substitution cipher, the position remains the same, but the actual character value changes

A $\longrightarrow$ D

Therefore, the relationship between the plaintext characters and ciphertext character is hidden.

Hence substitution mainly provides confusion.

Relationship between Transposition and Diffusion:

In transposition, the character itself doesn't change.

ABCDE $\longrightarrow$ CADBE

Therefore, the characters are moved to different positions. This spreads the information of the plaintext throughout the ciphertext.

Hence transposition mainly provides diffusion.

3) Confusion vs Diffusion

Analyse the following

Cipher Does it change

4) Transposition: Encryption

Encrypt the message "The arrow is pointing north direction" using

- Simple

with key = 4

- keyword

with keyword = ATLAS

- double

key 1: Break BREAK

key 2: Log

Simple →

T H E A
R R O W

I S P O
I N T I

N G N O
R T H D
I R E C
T I O N

Ciphertext

TRIINRITHRSNGTRIEOPTN
#EOAWOIODCN

key word: ATLAS

A T L A S

1 5 3 2 4

Cipher text:

T H E A R

TRPIRROAINNDTXEWIGHCX

R O W I S

- RSTOIIHXHOONTEN

P O I N T

I N G N O

R T H D I

R E C T I

O N X X X

Double: BREAK

LOG

BREAK

2 5 3 1 4

T H E A R

R O W I S

P O I N T

I N G N O

R T H D I

R E C T I

O N X X X

Cipher text:

AINNDTXTRPIRROEWIGHCX

RSTOIIHXHOONTEN

~~to~~ L O G

2 3 1

A I N

N D T

T R P

I R R

O E W

I G H

C R S

T O I

I H O

O N T

E N X

Ciphertext:

~~NTRRWHSTOTXIDRREGRO-~~

~~HNAN~~

Ciphertext: NTRRWHSTOTXANTIOICTIOEIDRREGROHN
NX.

Key word $\rightarrow$ SECRET

Total = 36

Key length $\rightarrow$ 6

ILED TNHPTR SOIIT MAXSA

XERXT ANONI SNFOTX

S Z I 4 3 6

S E C R E T

T H I S I S

A P L A I N

N T E X T

O R D E M O

N S T R A T

I O N X X X

PT:- ~~This is a plain text~~

THIS IS A PLAINN

TEXT FOR DEMONSTR-
-ATION

//_

Decryption:-

Simple ~~transit~~ transposition with key = 10

Ciphertext:-

TIDIE MRNME REOGO SANOW RSLTP EEEDO
SSSNU AMTUD BIENP GODAE PEIXD ELNSX

Total no. of char = 60

key = 10

No. of columns = 10

No. of rows = $\frac{60}{10} = 6$

T	I	D	I	E	M				
R	N	M	E	R	E				
O	G	O	S	A	N				
O	W	R	S	L	T				
P	E	E	E	D	O				
S	S	S	N	U	A				
H	T	U	D	B	I				
E	N	P	G	O	D				
A	E	P	E	I	X				
D	E	L	N	S	X				

TROOPS HEADING WEST NEED MORE SUPPLIES
SEND GENERAL DUBOIS MEN TO AID

5) Permutation
Find the inverse permutation for

(i) 4 3 1 5 2
 $P = (4, 3, 1, 5, 2)$

1 is in position 3

Similarly, $2 \rightarrow 5$
 $3 \rightarrow 2$
 $4 \rightarrow 1$
 $5 \rightarrow 4$

Inverse permutation = $(3, 5, 2, 1, 4)$

(ii) 7 4 1 3 5 2 6

$P = (7, 4, 1, 3, 5, 2, 6)$

$1 \rightarrow 3$ $5 \rightarrow 5$
 $2 \rightarrow 4$ $6 \rightarrow 7$
 $3 \rightarrow 1$ $7 \rightarrow 1$

$4 \rightarrow 2$ $P^{-1} = (3, 4, 1, 2, 5, 7, 1)$

(iii) 4 1 3 2

$P = (4, 1, 3, 2)$

$1 \rightarrow 2$ $4 \rightarrow 1$

$2 \rightarrow 4$

$3 \rightarrow 3$

$P^{-1} = (2, 4, 3, 1)$

(iv) 3 5 1 4 2

$$p = (3, 5, 1, 4, 2)$$

$$1 \rightarrow 3$$

$$4 \rightarrow 4$$

$$2 \rightarrow 5$$

$$5 \rightarrow 2$$

$$3 \rightarrow 1$$

$$p^{-1} = (3, 5, 1, 4, 2)$$

$$p^{-1} = (3, 5, 1, 4, 2)$$